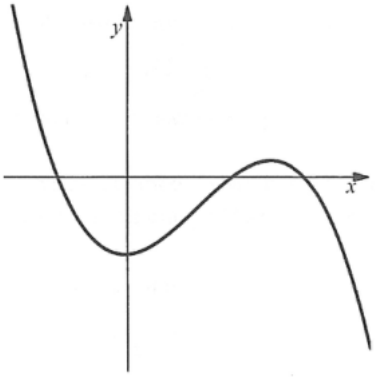
Differentiation

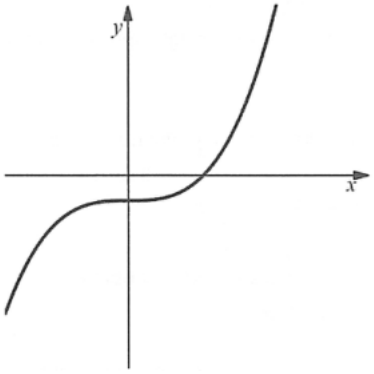
Questions 1–185

2.20 — Question 1. For each of the following graphs, identify the points of inflection on a copy of the graph and describe the change in concavity at each point of inflection:

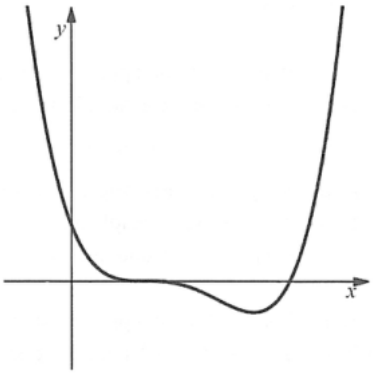
a)



b)



c)



2.20 — Question 2. Sketch a graph with a single point of inflection at $(0, 0)$ which is non-stationary, concave (concave downwards) before $(0, 0)$ and convex (concave upwards) after $(0, 0)$

2.20 — Question 3. Sketch a graph which has a stationary point which is a minimum at $x = -2$, a non-stationary point of inflection at $x = 1$ and a stationary point which is maximum at $x = 4$. Label each of the points clearly.

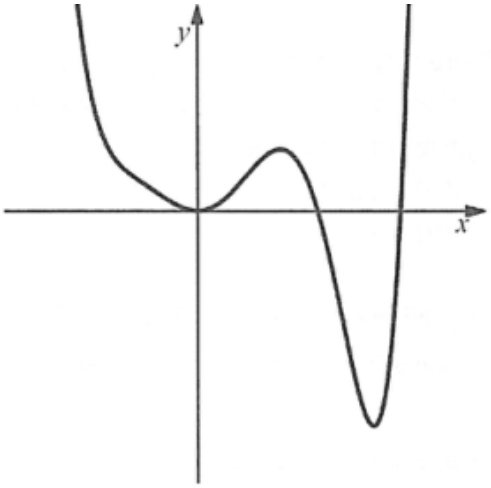
2.20 — Question 4. Determine whether each of the curves are convex (concave upwards), concave (concave downwards) or neither at the given point:

a) $y = 3x^3 - x^2 - x - 1$ at $x = 0$
b) $y = x^4 + 3x^2 + x - 6$ at $x = 1$
c) $y = 5x^3 - x^2 + x$ at $x = -1$
d) $y = x^3 - 12x^2 + 48x - 2$ at $x = 4$

2.20 — Question 5. The curve

$$y = x^6 - 3x^5 - 10x^4 + 10x^3 + 45x^2$$

is shown in the graph below.



Find all the points of inflection of the curve.

2.20 — Question 6. Determine the values of x for which

$$y = 12x^5 - 20x^4 - 50x^3 + 90x^2$$

is convex (concave upwards). Give your answer in set notation.

2.20 — Question 7. Find the coordinates of the points of inflection for each of the following curves. Determine in each case whether they are stationary or not.

a) $\mathrm{f}(x) = x^4 + 9x^3 - 15x^2 + 4$

b) $\mathrm{f}(x) = x^4 + 2x^3 - 2x - 1$

2.20 — Question 8. Given that the function

$$h = 2t^3 - 9t^2 + 12t - 3$$

is used to model the height, h metres, of a passenger on a ride over time t, find the value of t when $\frac{d^2h}{dt^2} = 0$ and state the physical relevance of this point.

2.20 — Question 9. The value of some shares £V during the first 12 months of the year may be modelled by

$$V = 0.25t^3 - 3.25t^2 + 10t + 20$$

where t is the number of months since the start of the year.

a) Use this model to predict the value of the shares when 12 months have passed.

b) Find when $\frac{d^2V}{dt^2} = 0$ and state its physical relevance.

2.20 — Question 10. Show that

$$y = 2x^3 + 16x^2 + 43x + 38$$

has a single non-stationary point of inflection.

2.20 — Question 11.

$$\mathrm{f}(x) = x^3 + 3x^2 + 3x + 2$$

a) Find the coordinates of the single stationary point of $y = \mathrm{f}(x)$
b) Show that the stationary point is a point of inflection of $y = \mathrm{f}(x)$
c) Hence, sketch the curve $y = \mathrm{f}(x)$

2.20 — Question 12.

$$\mathrm{f}(x) = x^2 - 3x - \frac{1}{x}, \qquad x \neq 0$$

a) Find the stationary points of $y = \mathrm{f}(x)$ and determine their nature.
b) Show that there are no non-stationary points of inflection on the curve $y = \mathrm{f}(x)$

2.20 — Question 13.

$$f(x) = 3x^4 - 4x^3 + 2$$

a) Show that the graph of $y = f(x)$ only has two stationary points.
b) Show that the two stationary points of $y = f(x)$ are different types of points.
c) Hence, sketch the curve $y = f(x)$

2.20 — Question 14. For the curve $y = 2x^4 + 4x^3$

a) Find the stationary points on the curve and determine their type.
b) Find the non-stationary point of inflection.
c) Hence, sketch the curve.

2.20 — Question 15. Find the values of a, b and c such that

$$y = x^3 + ax^2 + bx + c$$

has a single stationary point of inflection at the point $(5, 2)$

2.20 — Question 16. Find the values of a, b and c such that

$$y = -x^3 + ax^2 + bx + c$$

has a single stationary point of inflection at the point $(1, 6)$

2.20 — Question 17.

Prerequisite knowledge for Q17–Q18: Differentiation of the form ae^{kx} and solving equations involving e^x

Determine whether each of the curves are convex (concave upwards), concave (concave downwards) at the given point.

a) $y = e^x - x$ at $x = 2$
b) $y = 2e^x - e^{2x}$ at $x = 0$
c) $y = e^{2x} + 4$ at $x = -1$
d) $y = e^{-x} + x^2$ at $x = -1$
e) $y = e^{-x} - x^2$ at $x = 0$

2.20 — Question 18.

a) Show that

$$y = e^x - x^2$$

has a single non-stationary point of inflection.
b) Find the coordinates of this point.

2.20 — Question 19.

Prerequisite knowledge for Q19: Differentiation of Standard Functions

Show that there is a single non-stationary point of inflection on the curve

a) $y = x + \sin\left(\frac{1}{2}x\right), \qquad 0 < x \leq 2\pi$

b) $y = x^2 + \ln x, \qquad x > 0$

2.20 — Question 20.

Prerequisite knowledge for Q20: Standard Functions and Product Rule

The price per litre (in pence), P, from January to June in 2024 is modelled by function

$$P = 144 + (t-3)e^{0.5t}$$

where t is the number of months after 1 January 2024.

a) According to the model, find the minimum price of the fuel in this time frame and the month in which it occurs.

b) Show that the curve has no points of inflection and explain what this means in context.

2.21 — Question 1. Differentiation of $(ax+b)^n$ Differentiate each of the following:

a) $y = (x+3)^5$
b) $y = (4x+1)^6$
c) $y = (6-5x)^4$
d) $y = (2x-3)^7$
e) $y = (7-4x)^{\frac{1}{4}}$
f) $y = \frac{1}{x+2}$
g) $y = \frac{3}{(x-5)^2}$
h) $y = \frac{1}{(8-3x)^4}$
i) $y = \sqrt{2+x}$
j) $y = \frac{1}{\sqrt{3x-1}}$

2.21 — Question 2. Find $f''(x)$ for each of the following:

a) $f(x) = (7x-1)^5$
b) $f(x) = \sqrt{3-x}$
c) $f(x) = \frac{1}{\sqrt[3]{2x+9}}$

2.21 — Question 3. Find coordinates of any stationary points on the following curves:

a) $y = (2x - 7)^5$

b) $y = 4x + \frac{1}{x-1}$

2.21 — Question 4. It is given that $f(x) = (3x + 1)^4$.

a) Find $f'(x)$ and expand the result fully.

b) Expand $f(x)$ using binomial expansion, and hence find $f'(x)$ to confirm your answer to part a).

2.21 — Question 5. Find the tangent line to the curve $y = \sqrt{x+7}$ when $x = 2$ giving your answer in the form $ax + by = c$ where a, b and c are integers to be found.

2.21 — Question 6. The height of water, h cm in a harbour for a 12-hour period can be modelled by the equation

$$h = 16(t-6)^2 + 200, \qquad 0 \leq t \leq 12$$

and t is the time in hours after 6 am.

a) By using this model and calculus, predict the minimum height of the water which occurred that day.

b) By considering $\frac{d^2h}{dt^2}$, confirm that your answer to a) is a local minimum.

2.21 — Question 7. Differentiation of $\sin kx$ and $\cos kx$ Differentiate each of the following:

a) $y = \sin 3x$
b) $y = \cos 5x$
c) $y = -7\sin 2x$
d) $y = \frac{1}{2}\cos 6x$
e) $y = -\cos \frac{1}{4}x$
f) $y = -\frac{1}{5}\sin 10x$
g) $y = -\cos \pi x$
h) $y = \cos \frac{x}{\pi}$

2.21 — Question 8. Find the equation of the tangent to the curve $y = 2\sin 6x$ when $x = \pi$ giving your answer in the form $y = mx + c$.

2.21 — Question 9. Find the equation for the tangent to the curve $y = 1 + \cos 3x$ when $x = \frac{\pi}{2}$ giving your answer in the form $y = mx + c$.

2.21 — Question 10. Find the equation for the normal to the curve $y = 3 - \sin(\pi x)$ when $x = 2$ giving your answer in the form $y = mx + c$.

2.21 — Question 11. Given that $y = \cos 3x$, show that $\frac{d^2y}{dx^2} = -9y$.

2.21 — Question 12. Show that the curve $y = \cos x - x$ has a single stationary point in the range $0 < x < 2\pi$. Find the exact coordinates of the stationary point.

2.21 — Question 13. The number of visitors at a tourist attraction, p, may be modelled by the function

$$p = 300 - 40\cos\frac{\pi}{6}t$$

where t is the number of months since 1 January 2024.

a) Find the number of visitors predicted on 1 January.
b) Find an expression for $\frac{dp}{dt}$.
c) Find the value of $\frac{dp}{dt}$ when $t = 6$. What can you deduce in context from this value?
d) Find an expression for $\frac{d^2p}{dt^2}$. What can you deduce from its value when $t = 6$?

2.21 — Question 14. The distance, s cm, of a weight dropped from an equilibrium position may be modelled by the equation

$$s = 3\cos\frac{\pi}{3}t$$

where t is the number of seconds after the weight is released.

a) Find an expression for $\frac{ds}{dt}$.
b) Find the maximum value of $\frac{ds}{dt}$ giving your answer in exact form and find the first time when this occurs. Explain the relevance of this point in context.

2.21 — Question 15. Given the gradient of the normal to the curve $y = 6\cos 2\pi x$, $0 < x < \frac{\pi}{3}$ is $\frac{1}{6\pi}$, find the possible values of x for when this occurs giving your answer in exact form.

2.21 — Question 16. Differentiate $y = \sin x°$, giving your answer in degrees.

2.21 — Question 17. Differentiation of e^{kx}, $\ln kx$, a^x Differentiate each of the following:

a) $y = \ln 5x$
b) $y = 3^x$
c) $y = e^{5x}$
d) $y = 2\ln \frac{1}{4}x$
e) $y = -6^x$
f) $y = (e^x - 1)^2$

2.21 — Question 18. For each curve, find the coordinates of the stationary points and identify their type:

a) $y = 2e^x - 4x$
b) $y = \ln x - 8x$
c) $y = 2x^2 + 19x - 5\ln x$

2.21 — Question 19. Find the x-value for which $f'(x) = 1$ giving your answers in exact form:

a) $f(x) = \ln 2x$

b) $f(x) = e^{2x} + 4$

c) $f(x) = 3^x - 5$

2.21 — Question 20. Prerequisite knowledge for Q20: Points of Inflection Show that the curve $y = 8\ln x - 5x^2$ is concave (concave downwards) for all values of $x > 0$.

2.21 — Question 21. Find the equation of the tangent to the curve $y = \ln 5x$ when $x = \frac{e}{5}$ giving your answer in the form $y = kx$ where k is a constant.

2.21 — Question 22. Find the equation of the normal to the curve $y = 5^x$ when $x = 2$ in the form $ay + bx = c$ where a, b and c are constants given in exact form.

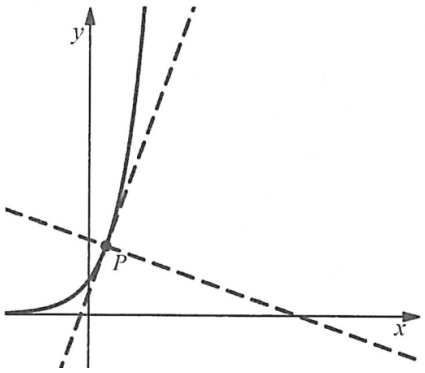
2.21 — Question 23. A population P of bugs has been found and a treatment applied so their population is reduced. The number of bugs after t days can be modelled by

$$P = 500 \times 20^{-\frac{1}{2}t}$$

a) Determine how many whole hours need to have elapsed for the number of bugs to drop to 50.
b) Determine the rate of change of bugs after 7 days.

2.21 — Question 24. Find the coordinates of the single stationary point of the curve $y = 9^x - 3^x$ and determine its type.

2.21 — Question 25. The graph below shows the curve with equation $y = 2 \times 2^x$.



The point P lies on the curve at $x = 1$. By finding the equation of the tangent and normal line at the point P, find the area of the triangle bounded by these two lines and the x-axis. Give your answer in the form

$$p \times \frac{1}{\ln 2} + q \ln 2$$

where p and q are constants to be found.

2.21 — Question 26. Show that the equation of the tangent to the curve $y = \ln x + a$ when $x = e^b$ is

$$e^b y - x = (a + b - 1)e^b$$

for constants a and b.

2.21 — Question 27. Differentiation of $\tan kx$ Differentiate each of the following:

a) $y = \tan 3x$
b) $y = 2\tan 4x$
c) $y = -\frac{1}{3}\tan 9x$
d) $y = -\tan \pi x$
e) $y = -\frac{1}{12}\tan \frac{6}{\pi}x$

2.21 — Question 28. Find the equation for the tangent to the curve $y = 4 + 2\tan \frac{\pi}{9}x$ when $x = \frac{9}{4}$ in the form $y = mx + c$.

2.21 — Question 29. Show that the equation for the normal to the curve $y = x - 2\tan\frac{\pi}{4}x$ when $x = 4$ crosses the x-axis at $x = 8 - 2\pi$.

2.21 — Question 30. Mix of Standard Functions Find the gradient function of each of the following:

a) $y = 3\sin 7x + 2\tan 2x$
b) $y = 15^x - 3\cos\frac{5}{2}x$
c) $y = \ln 180x + \tan\sqrt{2}x$
d) $y = \sqrt{x} - 3\sin 2x$
e) $y = e^{3x} + \sin 8x$
f) $y = \cos\frac{1}{4}x - \cos 2\pi x$
g) $y = 4^x + \tan \pi x$

2.21 — Question 31. Find $\frac{d^2y}{dx^2}$ for each of the following:

a) $y = 3x^5 - e^{5x}$
b) $y = x^2 - \sin 2x$
c) $y = \sqrt{x} + \ln 2x$
d) $y = \frac{1}{x} - \cos \frac{1}{2}x$
e) $y = e^{2x} - \sin \pi x$
f) $y = e^{-\frac{1}{2}x} + \sin 3x - \cos 3x$
g) $y = \frac{1}{\sqrt{x}} - \ln \frac{1}{2}x$

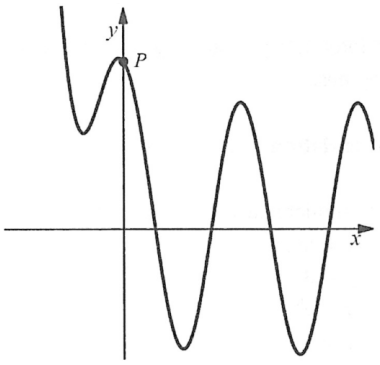
2.21 — Question 32. Find the exact value of $f'(x)$ at the given value of x, giving your answer as an exact value:

a) $f(x) = e^{2x} + x^{\frac{1}{4}}$ when $x = 1$
b) $f(x) = 5 \ln x + \frac{1}{x^2}$ when $x = 2$
c) $f(x) = 2^x + \sin 2\pi x$ when $x = \frac{1}{2}$
d) $f(x) = \cos(\frac{\pi}{2}x) + \tan(-\frac{\pi}{2}x)$ when $x = 0$

2.21 — Question 33. Determine whether the curve $y = e^{-2x} - 3\sin 5x$ is increasing or decreasing when $x = \pi$.

2.21 — Question 34. Find the equation for the tangent to the curve $y = \ln 2x + \sin \pi x$, $x > 0$ in the form $y = mx + c$ when $x = 3$.

2.21 — Question 35. The curve shown below is $y = e^{-2x} + 3\cos \pi x$.



Point P is where the curve meets the y-axis.

a) Find the tangent line to the curve at P.
b) Find the normal line to the curve at P.
c) Find the area bounded by the normal and tangent lines at point P and the x-axis.

2.21 — Question 36. Prerequisite knowledge for Q36: Points of Inflection Show that the curve $y = 2^x + \cos 3x$

a) is convex (concave upwards) at $x = \pi$
b) is concave (concave downwards) at $x = \frac{2\pi}{3}$

2.21 — Question 37. Show clearly that the curve $y = \sin 3x + \tan 3x$ has a single stationary point for $0 < x < \pi$ and state the coordinate of this stationary point.

2.22 — Question 1. Differentiation of $(ax^k + \ldots)^n$ Find the gradient function of:

a) $y = (2x^2 + 1)^3$
b) $y = (2x^3 - 5x)^4$
c) $y = (3x^4 - 1)^{1/2}$
d) $y = (1 - x^2)^{-3/2}$
e) $y = \frac{1}{5+\sqrt{x}}$
f) $y = (\frac{1}{x} - x)^7$

2.22 — Question 2. Find the gradient of the curve $y = \sqrt{5x^2 - 2x + 1}$ at $(2, \sqrt{17})$

2.22 — Question 3. Find the equation of the tangent to the curve $y = \sqrt{1 + 7x^2}$ when $x = 3$ giving your answer in the form $ay + bx = c$ where a, b and c are integers to be found.

2.22 — Question 4. Find the equation of the normal to the curve $y = \frac{1}{\sqrt{4-3x^2}}$ when $x = -1$ giving your answer in the form $ay + bx = c$ where a, b and c are integers to be found.

2.22 — Question 5. Find the coordinates of all the stationary points of each of the following curves:

a) $y = (3x^2 - 1)^4$

b) $y = \frac{1}{\sqrt{1+x^2}}$

2.22 — Question 6. For each of the following, find $\frac{dy}{dx}$ giving your answer in terms of y:

a) $x = 3y^2 - y$

b) $x = 2y^4 - 1$

c) $x = (3y^2 - 1)^{1/4}$

2.22 — Question 7. Find $\frac{dy}{dx}$ of each of the following at the given point:

a) $x = (3 - y)^5$ when $y = 2$

b) $x = (y^2 - 2y + 1)^{1/2}$ when $y = 4$

c) $x = \frac{1}{\sqrt{y^4+3}}$ for $y > 0$ when $x = \frac{1}{2}$

2.22 — Question 8. Find the tangent of the curve $y = \frac{1}{3}(x^2 + 5)^{3/2}$ at $x = 2$ giving your answers in the form $y = mx + c$

2.22 — Question 9. A curve has equation $x = (y - 2)^6$

a) Find $\frac{dx}{dy}$ in terms of x

b) Hence find $\frac{dy}{dx}$

2.22 — Question 10. Prerequisite knowledge for Q10: Points of Inflection Find the x-value of the point of inflection on the curve $\mathrm{f}(x)$ given that $\mathrm{f}'(x) = \frac{1}{(1+3x-2x^2)^3}$

2.22 — Question 11. Differentiation of $\sin(\mathrm{f}(x))$ **&** $\cos(\mathrm{f}(x))$ Differentiate each of the following:

a) $y = \sin^2 x$
b) $y = \cos^2 x$
c) $y = 2\sin(3x - 7)$
d) $y = -\frac{1}{5}\cos(\pi x - \frac{\pi}{2})$
e) $y = \cos\sqrt{x}$
f) $y = -2\sin(x^3)$
g) $y = \cos(\sin x)$
h) $y = \sin(\sin x)$
i) $y = \sqrt{\sin x}$
j) $y = \sqrt[3]{\cos x}$
k) $y = 1 - \cos^3(5x)$

2.22 — Question 12. Show that $\frac{d}{dx}\left[\frac{1}{\sin x}\right] = -\operatorname{cosec} x \cot x$

2.22 — Question 13. A model for the tide height over a 24-hour period in Southampton is

$$x = 2.85 + 0.75 \sin\left(\frac{17\pi}{120}t + \frac{3\pi}{2}\right)$$

where the depth of the water is x metres and t is the number of hours passed since midnight.

a) State the maximum and minimum height the water reaches.

b) By using Calculus, find the two times at which it reaches its maximum depth in the first 24-hour period.

2.22 — Question 14. A curve is given by $y = a\sin^2 x + b\cos^2 x$ where a and b are positive constants and are such that $a \neq b$. Show that the gradient function of the curve at $x = \frac{\pi}{4}$ is never zero.

2.22 — Question 15. A model for the number of daylight hours, t (in hours), on the xth day of the year may be given by

$$t = 10 + 2.7\sin\left(\frac{2\pi}{365}(x - 78)\right)$$

Use this model to compare how the number of hours of daylight is changing on day 17 compared to on day 85.

2.22 — Question 16. Show that the stationary points of $f(x) = \sin(\frac{\pi}{2}e^x)$ occur when $x = \ln k$ where k is a positive odd integer.

2.22 — Question 17. Show that $f(x) = \frac{1}{2}x + \cos(\frac{2}{x})$ is an increasing function for all $x > 2$

2.22 — Question 18. Differentiation of $e^{f(x)}$, $\ln(f(x))$, $a^{f(x)}$ Differentiate each of the following:

a) $y = (e^{2x} - 1)^5$
b) $y = (5^x - 1)^4$
c) $y = (\ln x)^4$
d) $y = e^{3x^2-1}$
e) $y = 4^{x^2+5x+1}$
f) $y = -5^{2x^2+1}$

2.22 — Question 19. Find coordinates of the stationary points on the following curves:

a) $y = 6x - e^{2x}$
b) $y = 2\ln(x - x^2)$

2.22 — Question 20. Differentiate each of these:

a) $y = \ln(2x^3 + x^2 - 1)$
b) $y = \ln(\sqrt{x})$
c) $y = \ln(\frac{1}{\sqrt{x}})$
d) $y = \ln(\sin x)$
e) $y = \ln(e^x + 1)$
f) $y = \ln(\ln x)$

2.22 — Question 21. Differentiation of $\tan(\mathrm{f}(x))$ Differentiate each of the following:

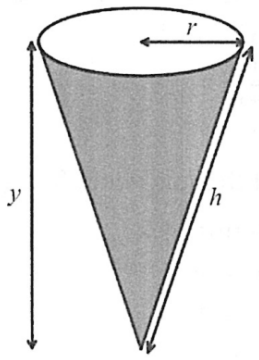
a) $y = \tan\sqrt{x}$
b) $y = \tan(\sin x - x^2)$
c) $y = e^{\tan x}$
d) $y = \tan(\ln x + e^{2x})$
e) $y = \tan(\tan x)$

2.22 — Question 22. Show that $\frac{d}{dx}[\ln(\tan x)] = \operatorname{cosec} x \sec x$

2.22 — Question 23. A curve is given by $y = \tan(x^2)$ where $0 < x < \sqrt{\frac{\pi}{2}}$. Show that the curve is always increasing.

2.22 — Question 24. A curve is given by $y = \tan^2(ax)$ where $0 < a \leq 2\pi$. Given that when $x = 1.5$, $\frac{dy}{dx} = 0$ find the possible values of a

2.22 — Question 25. An ice cream cone is to be made from 30π cm^2 of wafer. The ice cream cone may be modelled with radius r cm, height y cm and slant height h cm and an assumption of negligible thickness.



Curved surface area: $A = \pi rh$
Volume: $V = \frac{1}{3}\pi r^2 y$

a) Show that the volume is given by $V = \frac{1}{3}\pi\sqrt{900r^2 - r^6}$
b) Hence, find the maximum volume of the cone.

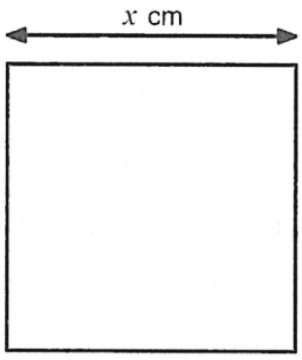
2.23 — Question 1. State each of the following:

a) The area of a circle in terms of its radius r
b) The area of an equilateral triangle in terms of its side length x
c) The volume of a cylinder V in terms of its radius r and its height h
d) The volume of a sphere in terms of its radius r
e) The surface area of a sphere in terms of its radius r
f) The volume of a cone with base of radius r and height h in terms of r and h

2.23 — Question 2. Given

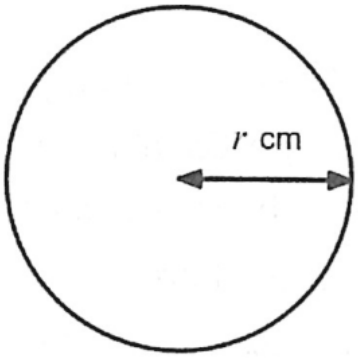
a) $A = \frac{1}{2}\pi r^2$ and $\frac{dr}{dt} = 2$, find $\frac{dA}{dt}$ in terms of r
b) $y = 10x^2$ and $\frac{dx}{dt} = -\frac{1}{2}$, find $\frac{dy}{dt}$ in terms of x
c) $P = 2x^3 - x$ and $\frac{dx}{dt} = 3$, find $\frac{dP}{dt}$ in terms of x
d) $M = -2x^5 + x^2$ and $\frac{dx}{dt} = \frac{1}{2}$, find $\frac{dM}{dt}$ in terms of x

2.23 — Question 3. The side of a square x is increasing at a rate of 10 $\mathrm{cm\,s^{-1}}$.



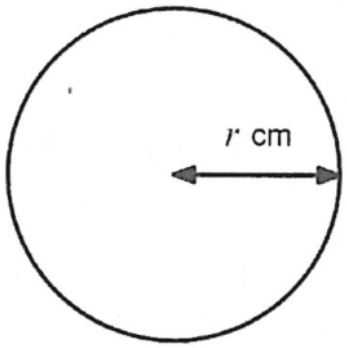
Find the rate at which the area A of the square is increasing when the side of the square is exactly 4 cm.

2.23 — Question 4. The radius of a circle r is increasing at a rate of 3 $\mathrm{cm\,s^{-1}}$.



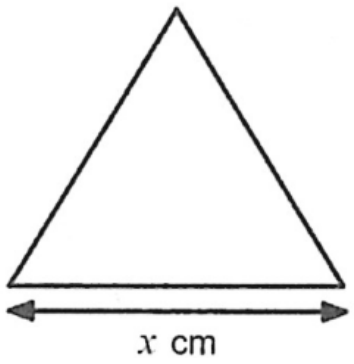
Find the rate at which the area A of the circle is increasing when the radius of the circle is 5 cm giving your answer in terms of π

2.23 — Question 5. The radius of a circle r is decreasing at a constant rate of 0.1 $\text{cm}\,\text{s}^{-1}$.



Find the rate at which the perimeter of the circle is decreasing, giving your answer in terms of π

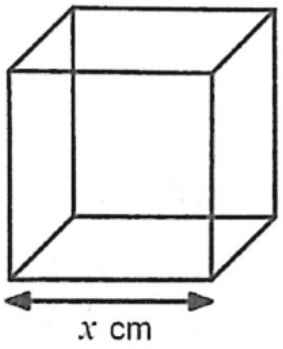
2.23 — Question 6. The side of an equilateral triangle x is increasing at a rate of 12 $\text{cm}\,\text{s}^{-1}$.



Find the rate at which the area of the triangle is increasing when the side of the triangle is $2\sqrt{3}$ cm.

2.23 — Question 7. The surface area of a sphere is increasing at a rate of 12π cm^2 s^{-1}. Find the rate of change of the radius of the sphere exactly when its diameter is 6 cm.

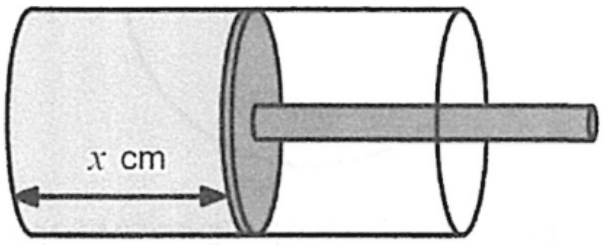
2.23 — Question 8. A cube is growing so that its volume increases at a rate of 5 cm^3 s^{-1}.



The length of each of its sides is x cm.

a) State the volume of the cube V and the surface area of the cube A in terms of x
b) Show that the rate of increase of the surface area of the cube may be given by $\frac{dA}{dt} = \frac{20}{x}$
c) Hence, find the rate of increase of the cube's surface area when x has a length of 4 cm.

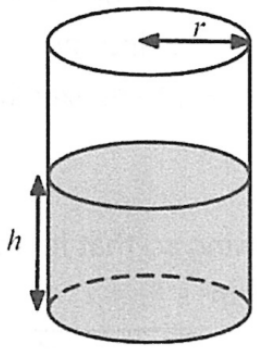
2.23 — Question 9. A piston can slide inside a combustion cylinder which is closed at one end. The cylinder is filled with gas whose pressure p in atmospheric units is inversely proportional to the distance of the piston from the closed end, x cm.



It is known that the atmospheric pressure is 1.2 when x is 50 cm. At a different point in time, the distance from the closed end is 25 cm and the piston is moving away from the closed end at 5 cm per minute.
Find the rate at which the pressure is changing at that point in time.

2.23 — Question 10. A sphere is growing so that its volume increases at a rate of 4 $\text{cm}^3\,\text{s}^{-1}$. Find the rate of increase of the sphere's surface area when its diameter is 50 cm.

2.23 — Question 11. A cylindrical tank with radius r cm has water being poured into it at a rate of 10 $\text{cm}^3\,\text{s}^{-1}$.



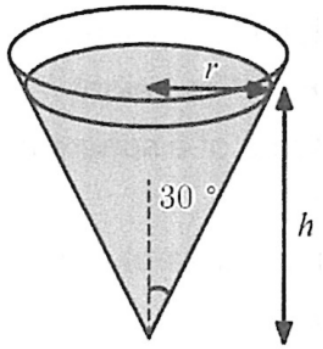
The height of water after t seconds is given by h cm.

a) Find an expression for the rate of increase of the water level in the tank in terms of r

b) Given the tank has diameter 90 cm, find the rate of change of the water level giving your answer to 3 significant figures.

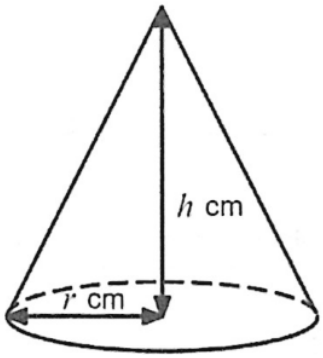
2.23 — Question 12. A cylindrical tank with radius 10 cm which initially starts full, has water dripping out of a hole in the bottom at a rate of 5 cm^3 per minute. The height of water after t minutes is given by h cm.
Find the rate of change of the water level giving your answer to 3 significant figures.

2.23 — Question 13. Water is being poured into an empty cylindrical cone of vertical height 10 cm at a constant rate of 6 $\text{cm}^3\,\text{s}^{-1}$. The cylindrical cone has an angle of 30° to the vertical.



a) Show that the volume of the water V when it is at height h cm is given by $V = \frac{1}{9}\pi h^3$
b) Find $\frac{dV}{dh}$
c) Find $\frac{dh}{dt}$ when the cone is full.

2.23 — Question 14. Sand is being poured onto a table and forms a pile in the shape of a cone. The height of the cone is equal to the diameter of the base.



The sand is poured at 3 cm^3 per second. Find the rate at which the height of the pile is increasing when the base has a diameter of 4 cm.

2.23 — Question 15. The volume of a cube with sides of length x metres is increasing at a constant rate. Show that the rate of increase of the sides, is inversely proportional to the area of one side of the cube.

2.23 — Question 16. The radius of a circle r changes at a rate which is inversely proportional to the square of r

a) Given that the rate is 0.5 $\text{cm}\,\text{s}^{-1}$ when the radius is 2 cm, find $\frac{dr}{dt}$

b) Show that the rate at which the area of the circle, A, changes satisfies $\frac{dA}{dt} = \sqrt{\frac{16\pi^3}{A}}$

2.24 — Question 1. Find $\frac{dy}{dx}$ for each of the following:

a) $y = x(x-1)^2$
b) $y = x^2(2x+7)^2$
c) $y = 4x(7-x)^4$
d) $y = (x+2)^2(3x+1)^3$
e) $y = (3-x)^2(2x+9)^3$
f) $y = \sqrt{2x+5}(4x+1)^2$
g) $y = (\frac{1}{x} - 2x)(\frac{1}{x^3} + \frac{2}{x^5})$

2.24 — Question 2. Three students attempt to find $\frac{d}{dx}((x^2-1)(x^3+5))$. Identify the correct version and spot the error(s) in the other two:

Student A:

$$(x^2-1)(x^3+5) = x^5 - x^3 + 5x^2 + 5$$

$$\frac{dy}{dx} = 5x^6 - 3x^2 + 10x$$

Student B:

$$\frac{dy}{dx} = (2x)(x^3+5) + (x^2-1)(3x^2) = 5x^4 - 3x^2 + 10x$$

Student C:

$$\frac{dy}{dx} = 2x \times 3x^2 = 6x^3$$

2.24 — Question 3. Differentiate the following:

a) $y = x \ln x$
b) $y = \sqrt{x} \tan x$
c) $y = \sin x \cos x$
d) $y = \cos x \ln x$
e) $y = 2^x \sin x$
f) $y = (x^2 + 1) \ln x$

2.24 — Question 4. Differentiate and fully factorise the following:

a) $y = xe^x$
b) $y = x^2 4^x$
c) $y = e^x \tan x$
d) $y = 2^x \sin x$
e) $y = (x^2 - 12)e^{-2x}$
f) $y = e^{4x} \sin 2x$

2.24 — Question 5. Find the equation of the tangent line to the curve at the given point, giving your answers in the form $y = mx + c$:

a) $y = x^3(x+2)^2$ at $x = 1$
b) $y = x^5\sqrt{3x+10}$ at $x = 2$
c) $y = (x^2 - x + 4)e^x$ at $x = 0$

2.24 — Question 6. Fully factorise each of the following:

a) $4x^2 + 2x^2(x+1)$
b) $3(2x-1)^5 - (x+1)(2x-1)^4$
c) $4(3x+1)^3 + (x+2)(3x+1)^2$
d) $\frac{1}{3}(6x+5)^3 + (x-1)(6x+5)^2$
e) $\frac{1}{2}(1-x)^{-5} + 4(1-x)^{-6}$
f) $4(4x+1)^3(3x-1) + 9(3x-1)^2(4x+1)^2$

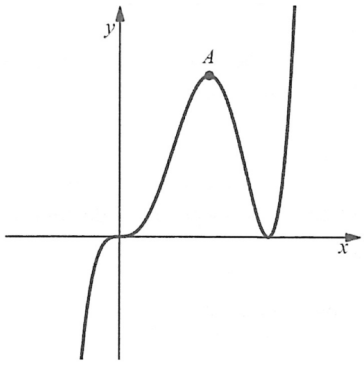
2.24 — Question 7. Solve each of the following:

a) $2(x+1)^3 + (x-1)(x+1)^2 = 0$

b) $\frac{1}{3}(2x+1)^{-2} + (2x+1)^{-3} = 0$

c) $5(7x-1)^3(x-1) - 3(x-1)^2(7x-1)^2 = 0$

2.24 — Question 8. The curve shown below is $y = x^3(3x-5)^2$.



A local maximum value is at the point A.

a) Find the coordinates of A.

b) State the coordinates of the other stationary points.

c) State the values of y for which the $y = x^3(3x-5)^2$ has three unique solutions.

2.24 — Question 9. For each of the following, differentiate and fully factorise:

a) $y = x(x+3)^7$

b) $y = (x-3)(2x+1)^4$

c) $y = 5x(2x+1)^4$

d) $y = (1+x^2)e^{2x}$

2.24 — Question 10. A curve is given by $y = 4000xe^{-2x}$.

a) Find the stationary point of the curve.

b) Determine its type.

2.24 — Question 11. The number of employees y, in a company in the first 12 months may be modelled by $y = 150 - 3t\sin(\frac{\pi}{6}t)$ where t is the number of months after 1st January.

a) Find the number of employees predicted by the model on 1st January.
b) Find an expression for $\frac{dy}{dt}$
c) Find the rate of change in the number of employees predicted by the model on 1st April.
d) Interpret your result in part c) giving your answer in context.

2.24 — Question 12. Find the equation of the tangent line to the curve $y = x^4e^{2x}$ when $x = 1$. Give your answer in the form $y + pe^2x + qe^2 = 0$

2.24 — Question 13. A curve is given by $y = x \ln x$, $x > 0$. Find the stationary point and determine its type.

2.24 — Question 14. A curve is given by $y = e^{-x} \sin x$, $x \geq 0$.

a) Show that the curve has a stationary point at $(\frac{\pi}{4}, \frac{\sqrt{2}}{2} e^{-\pi/4})$

b) Determine its type.

2.24 — Question 15. Prerequisite knowledge for Q15: Geometric Sequences A curve is given by $y = 10e^x \cos x$, $x \geq 0$.

a) Show that the x coordinates of the stationary points of the curve satisfy the equation $\tan x = 1$

The y-values of the stationary points satisfy a geometric sequence,

b) find the value of the common ratio r.

2.24 — Question 16. Find $\mathrm{f}'(x)$ for each of the following:

a) $\mathrm{f}(x) = e^{10x^2} \sin(2x^3)$

b) $\mathrm{f}(x) = 5^{x^2} \cos(1 - x^2)$

c) $\mathrm{f}(x) = (x^2 + 1)^3 \ln(x^2)$

2.24 — Question 17. Prerequisite knowledge for Q17: Points of Inflection Show that $y = 2^{3x^2}$ is convex (concave upwards) for all values of x

2.24 — Question 18. Show that the curve $y = (5 - 2x)\ln(x + 1)$ has a stationary point at $x \approx 1.04$

2.24 — Question 19. For each of the following curves find $\frac{dy}{dx}$, find the coordinates of any stationary points.

a) $y = x^2e^x$

b) $y = (3x-1)^4(3x^2+1)^5$

c) $y = \sqrt{x}(x-1)^2$, $x > 0$

2.24 — Question 20. For the curve given by $x = (y+2)\ln(3y-5)$, $y > \frac{5}{3}$

a) Find $\frac{dx}{dy}$ in terms of y

b) Hence find the value of $\frac{dy}{dx}$ when $x = 0$

2.24 — Question 21. A curve has the equation $y = x^2 \tan^3 x$, $-\frac{\pi}{2} < x < \frac{\pi}{2}$. Show that the gradient of the curve at $x = \frac{\pi}{4}$ is $\frac{\pi}{8}(3\pi + 4)$

2.24 — Question 22. The height, h metres, from the ground of a bungee jumper is modelled by the equation

$$h = 10 + 25e^{-0.05t}(1 + \cos(\tfrac{\pi}{3}t))$$

where t is the time in seconds after jumping.

a) State the height above the ground of the start position.
b) Find $\frac{dh}{dt}$ and the value of $\frac{dh}{dt}$ when $t = 3$
c) Explain what is happening to the bungee jumper at precisely 3 seconds.

2.24 — Question 23. Given that $f(x) = x\sqrt{4x+3}$, $x > -\frac{3}{4}$

a) Find $f'(x)$

b) Show that $f''(x) = \frac{12(x+1)}{(3+4x)^{3/2}}$

c) Find the x-value of the turning point and determine its nature.

2.25 — Question 1. Standard Functions Find $f'(x)$ and simplify:

a) $f(x) = \frac{3x+1}{2x-1}$

b) $f(x) = \frac{x}{1-4x}$

c) $f(x) = \frac{3x-3}{3x-1}$

d) $f(x) = \frac{e^x}{x-3}$

e) $f(x) = \frac{\ln x}{x^3}$

f) $f(x) = \frac{\sin(10x)}{2x^3}$

g) $f(x) = \frac{5^x}{x^2}$

h) $f(x) = \frac{e^{2x}}{\cos x}$

i) $f(x) = \frac{\ln 5x}{2x+1}$

j) $f(x) = \frac{x^2-1}{\sin x}$

2.25 — Question 2. Given that $y = \frac{x-5}{\sqrt{x}}$ where $x \neq 0$, show that

$$\frac{dy}{dx} = \frac{x+p}{qx\sqrt{x}}$$

where p and q are integers to be found.

2.25 — Question 3. Given that $y = \frac{3^x}{x^2 \ln 3}$ where $x \neq 0$, show that

$$\frac{dy}{dx} = \frac{3^x(x \ln 3 - k)}{x^3 \ln 3}$$

where k is an integer to be found.

2.25 — Question 4. Simplify each of the following:

a) $\frac{(x+1)^2-3x(x+1)}{(x+1)^4}$

b) $\frac{(x-4)^2-2(x+1)(x-4)}{(x-4)^4}$

c) $\frac{2(x+2)^3-4(7x+1)(x+2)^2}{(x+2)^6}$

d) $\frac{4(x+3)^3(5x-1)^3-3(x+3)^4(5x-1)^2}{(5x-1)^6}$

e) $\frac{-6(1-2x)^3(2x+5)^3+6(1-2x)^4(2x+5)^2}{(1-2x)^8}$

2.25 — Question 5. Show that

$$\frac{(x+1)^{\frac{1}{2}}e^x - \frac{1}{2}(x+1)^{-\frac{1}{2}}e^x}{x+1}$$

may be rearranged to

$$\frac{e^x(1+2x)}{2(x+1)^{\frac{3}{2}}}$$

2.25 — Question 6. Find $f'(x)$ and simplify:

a) $f(x) = \frac{4x}{(x+1)^2}$

b) $f(x) = \frac{9x-2}{(1+2x)^3}$

c) $f(x) = \frac{e^{4x}}{(x+1)^2}$

2.25 — Question 7. Find the x-value of the stationary points of the following:

a) $y = \frac{4x-35}{(2x+5)^4}$

b) $y = \frac{(3-2x)^2}{(1+x)^3}$

c) $y = \frac{e^{\frac{1}{2}x}}{2(x-2)^2}$

d) $y = \frac{e^{ax}}{(x+a)^2}, \quad a \neq 0$

2.25 — Question 8. Reciprocal Trigonometric Functions By writing each first as a quotient, show that:

a) $\frac{d}{dx}(\tan x) = \sec^2 x$

b) $\frac{d}{dx}(\operatorname{cosec} x) = -\operatorname{cosec} x \cot x$

c) $\frac{d}{dx}(\sec x) = \sec x \tan x$

2.25 — Question 9. Find $\frac{dy}{dx}$ for each of the following:

a) $y = \tan 3x$

b) $y = \operatorname{cosec} 2x$

c) $y = \sec \frac{1}{3}x$

d) $y = -2\tan \pi x$

e) $y = \tan(-\frac{1}{5}x)$

f) $y = -\sec 9x$

g) $y = \operatorname{cosec}(-\frac{1}{2}x)$

2.25 — Question 10. Mix of Functions Find when the curve $y = \frac{2x-1}{4x^2+3}$ is decreasing, giving your answer in interval notation.

2.25 — Question 11. Find any stationary points of the curve $y = \frac{2x-4}{(4x-7)^2}$ and determine their nature.

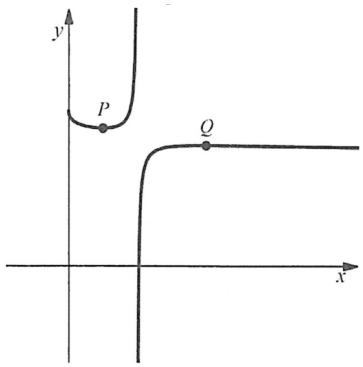
2.25 — Question 12. $f(x) = -\frac{1}{2}\sec \pi x$, $0 < x < 2$.

a) Find the stationary point on the curve $y = f(x)$.

b) Determine its type.

2.25 — Question 13. Prerequisite knowledge for Q13: Points of Inflection Show that the curve $y = \frac{\ln x}{x^2}$, $x > 0$ has exactly one single non-stationary point of inflection.

2.25 — Question 14. The curve $y = \frac{2\sqrt{x}+3x-9}{x-2}$, $x > 0$ is shown on the diagram below.



The stationary points are labelled P and Q.

a) Find the coordinates of P and Q.

b) Find the equation of the straight line passing through P and Q.

A triangle is bounded by the line passing through the points P and Q, the x-axis and the y-axis.

c) Find the area of the triangle.

2.25 — Question 15. Given $f(x) = \frac{ax-1}{x^2-a}$ show that the gradient at $x = 0$ is always -1.

2.25 — Question 16. A function is defined by $f(x) = \frac{3}{k+x^2}$, $x > 0$ and k is a positive constant.

a) Show that f is a decreasing function for all values of k.

The point of inflection on the curve is at $x = \frac{2}{\sqrt{3}}$.

b) Find the value of k.

2.25 — Question 17. Given that $f(x) = \frac{e^{-x}}{k+10x^2}$ where k is a positive constant,

a) find $f'(x)$.

b) Hence find the range of values of k such that $y = f(x)$ has at least one stationary point.

2.25 — Question 18. The Chain Rule Find $\frac{dy}{dx}$ for each of the following:

a) $y = \tan(x^2 - 1)$

b) $y = \text{cosec}(e^{2x})$

c) $y = \sec(\sin x)$

2.25 — Question 19. Show that the curve $y = \frac{e^{3x-7}}{(2x-3)^2}$ has exactly one stationary point.

2.25 — Question 20. $f(x) = \frac{(x^2+1)^2}{10(1-2x)}$, $x \neq k$.

a) State the value of k.

b) Show that the x-values of the maximum and minimum point on the curve $y = f(x)$ satisfy the equation $3x^2 - 2x - 1 = 0$.

c) Hence find the exact values of the maximum and minimum points.

d) Sketch $f(x)$.

2.25 — Question 21. Find $\frac{dy}{dx}$ for each of the following:

a) $y = \frac{e^{x^2}}{x^2-1}$

b) $y = \frac{\sin(x^3)}{e^{2x+1}}$

c) $y = \frac{3^{\cos x}}{x^2}$

2.25 — Question 22. It is given that $f(x) = (\frac{x^2-1}{x-3})^4$, where $x \neq 3$.

a) Find $\frac{d}{dx}(f(x))$.

b) Hence show that the tangent line to the curve $y = f(x)$ at the point $x = 2$ is $y - 756x + 1431 = 0$.

2.25 — Question 23. It is given that $f(x) = (\frac{\ln x}{x^2})^2$, where $x > 0$.

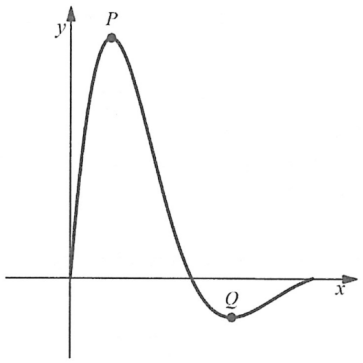
a) Find $\frac{d}{dx}(f(x))$.

b) Hence show that the curve $y = f(x)$ has a stationary point at $x = 1$.

2.25 — Question 24. The sketch below shows the curve

$$y = \frac{5\sin 3x}{e^{\sqrt{3}x+2}}$$

for $0 \leq x \leq \frac{2\pi}{3}$.



The stationary points are labelled P and Q.

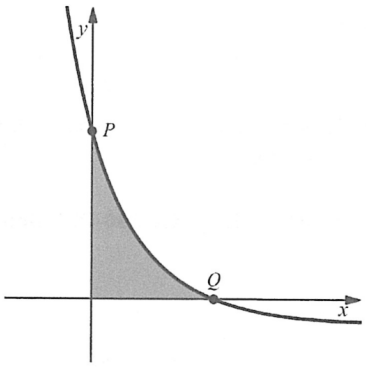
a) Show that the x coordinates of the stationary points are solutions to the equation $\tan 3x = \sqrt{3}$.

b) Find the x values of the stationary points of the curve.

2.25 — Question 25. $f(x) = \frac{\ln(x+1)}{(x+1)^2}$, $x \geq 0$.

a) Show that $f'(x) = \frac{1-2\ln(x+1)}{(x+1)^3}$.

The diagram below shows the area enclosed by the curve $y = \frac{1-2\ln(x+1)}{(x+1)^3}$ and the x-axis and y-axis.



b) Show that the shaded region is $\frac{1}{2e}$.

2.25 — Question 26. Prerequisite knowledge for Q26–Q29: The Product Rule Two functions

$$f(x) = (1-2x)(2x^2+ax+b) \qquad \text{and} \qquad g(x) = \frac{9(ax+b)}{x^2+2}$$

where a and b are integers. Given that $f'(1) = g'(1) = -8$ find the values of a and b.

2.25 — Question 27. Find $f'(x)$ for each of the following:

a) $f(x) = \frac{x^2e^x}{2x-1}$

b) $f(x) = \frac{2^x \sin x}{\sqrt{x}}$

c) $f(x) = \frac{x \ln x}{(x-1)^2}$

2.25 — Question 28. A curve has equation $y = \frac{(x^2-2x-3)e^x}{\sqrt{2x+1}}$. Show that the equation of the tangent to the curve at the point $x = 0$ is $y = -2x - 3$.

2.25 — Question 29. A curve is given by $y = \frac{e^{\frac{1}{2}x}}{x^2+2x+1}$ for $x \geq 0$.

a) Find the x-value of the single stationary point on the curve.

b) Show that $\frac{d^2y}{dx^2} = \frac{e^{\frac{1}{2}x}(x^2-6x+17)}{4(x+1)^4}$.

c) Hence show why the curve has no points of inflection.

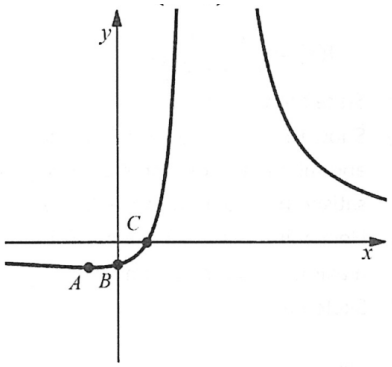
2.25 — Question 30. Prerequisite knowledge for Q30–Q31: Double Angle Formulae Show that $\frac{d}{dx}(\frac{\sin 2x}{\cos x}) = 2\cos x$.

2.25 — Question 31. Show that

$$\frac{d}{dx}(\frac{2\sin x}{5\sin x + 5\cos x}) = \frac{p}{q(1+\sin 2x)}$$

where p and q are integers and $\frac{p}{q}$ is in its most simplified form.

2.25 — Question 32. Prerequisite knowledge for Q32–Q34: Domain and Range of Functions The sketch below shows the curve $y = f(x)$ where $f(x) = \frac{x-1}{(x-3)^2}$, $x \neq k$.



The stationary point is labelled A. The intercepts with the axes are labelled B and Q.

a) State the value of k.
b) Find the coordinates of the points B and C.
c) Find the coordinates of A.
d) Sketch on separate axes, labelling the new values of A, B and C:
   i) $f(x) + 1$
   ii) $f(2x)$

2.25 — Question 33. Given that $f(x) = \frac{2\ln x - 5}{\ln x - 1}$.

a) State the maximum domain of f.
b) Show that $f(x)$ is an increasing function.
c) Find the set of values for which $f(x) < 0$.

2.25 — Question 34. Given $f(x) = \frac{2^x}{x-1}$.

a) State the maximum possible domain of f.

b) Find $f'(x)$ and hence find the coordinate of the single stationary point of f giving your answer to 3 significant figures.

c) Hence find the range of f.

2.26 — Question 1. Find

a) $\frac{d}{dx}(x + y^2)$

b) $\frac{d}{dx}(xy)$

c) $\frac{d}{dx}(x^2 y)$

d) $\frac{d}{dx}(e^x \sin y)$

e) $\frac{d}{dx}(e^y \cos x)$

f) $\frac{d}{dx}(x \ln y - 5)$

g) $\frac{d}{dx}(2^x y + 3)$

2.26 — Question 2. For each of the following, find $\frac{dy}{dx}$:

a) $y^2 = 8x$
b) $xy = 12$
c) $x^3 + xy^2 - y^2 = 0$
d) $\ln x + y^2 = 2x$
e) $5x^2 - 2xy + 5y^2 = 25$
f) $\sin x + xe^y = 1$
g) $x \cos y - 5 = x$
h) $x \ln y + y \ln x = 10$

2.26 — Question 3. By using implicit differentiation, find the tangent line to the circle $x^2 + y^2 = 25$ at the point $(4, 3)$.

2.26 — Question 4. Use implicit differentiation on $e^y = x$ to show that $\frac{d}{dx}(\ln x) = \frac{1}{x}$.

2.26 — Question 5. Use implicit differentiation on $\ln y = \ln a^x$ to show that $\frac{d}{dx}(a^x) = a^x \ln a$.

2.26 — Question 6. Differentiation of Inverse Trigonometric Functions Use implicit differentiation on $\sin y = x$ to show that $\frac{d}{dx}(\arcsin x) = \frac{1}{\sqrt{1-x^2}}$.

2.26 — Question 7. Use implicit differentiation on $\cos y = x$ to show that $\frac{d}{dx}(\arccos x) = -\frac{1}{\sqrt{1-x^2}}$.

2.26 — Question 8. Use implicit differentiation on $\tan y = x$ to show that $\frac{d}{dx}(\arctan x) = \frac{1}{1+x^2}$.

2.26 — Question 9. Mix Find the value(s) of $\frac{dy}{dx}$ at the given point:

a) $\sin 2y - x = 2$ at $(-2, \pi)$

b) $x^2 y = x$ when $x = 2$

c) $xe^y - x^2 + 4 = 0$ at $y = \ln 3$

2.26 — Question 10. Show that if $x^2 \sin y = 4$ then $\frac{dy}{dx} = -\frac{8}{x\sqrt{x^4-16}}$.

2.26 — Question 11. Show that if $x^2 \tan y = 1$ then $\frac{dy}{dx} = -\frac{2x}{x^4+1}$.

2.26 — Question 12. The equation of a curve is given by $x^2 + 2y^3 = 2$.

a) Find the coordinates of the stationary points on the curve.

b) Find the coordinates of the points where the curve has an infinite gradient.

2.26 — Question 13. Prerequisite knowledge for Q13: Inverse Functions

a) Sketch the graphs of $y = f(x) = x^2 + 4$ for $x \geq 0$ and its inverse $y = g(x)$ on the same axes, labelling any intersections with the axes.

b) Show that $f'(3) = \frac{1}{g'(13)}$.

2.26 — Question 14. Given that $f(x) = x(x-3)$, $x \geq 3$, calculate the gradient of the graph $y = f^{-1}(x)$ at the point $(10, 5)$.

2.26 — Question 15. The equation of a curve is given by $(x-2)^2 + 2 = e^y$. Find the coordinates of the point where the tangent line is parallel to the x-axis.

2.26 — Question 16. Show that the curve $xy^2 + \sqrt{y} = 5$, $y > 0$ has no stationary points.

2.26 — Question 17. A curve has equation $(2x + y)^2 = x - y$.

a) Find the tangent lines to the curve at $x = \frac{1}{4}$.

b) Hence, show that the two tangent lines meet at the point $(-\frac{5}{12}, \frac{1}{3})$.

c) Find the exact area of the triangle bound by the two tangent lines and the x-axis.

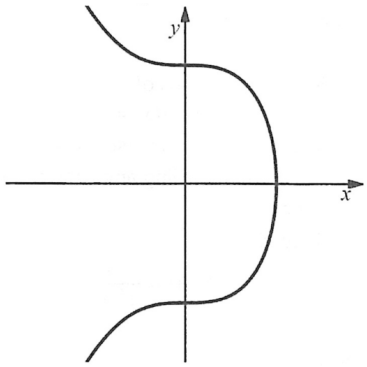
2.26 — Question 18. Show that for the curve $5x^2 + 12xy - 4y^2 = 10$ there are no tangent lines parallel to the line $y = \frac{3}{2}x$.

2.26 — Question 19. A curve has equation $(x + y)^3 = 10x^2 - 2y$.

a) Show that the point $(1, 1)$ lies on the curve.

b) Show that the tangent to the curve at the point $(1, 1)$ is given by $7y - 4x = 3$.

2.26 — Question 20. The curve $x^3 + y^2 = 5$ is shown in the diagram below.



a) Show that the point $(1, 2)$ lies on the curve.
b) Find the normal line to the curve at the point $(1, 2)$ in the form $ax + by = c$ where a, b and c are integers to be found.
c) Show that the normal line found in part b) does not intersect the equation again.

2.26 — Question 21. The equation of a curve is given by $x^2 \sin y + \cos y = \frac{1}{2}x$.
a) Show that $(0, \frac{3\pi}{2})$ lies on the curve.
b) Show that $\frac{dy}{dx} = \frac{4x \sin y - 1}{2 \sin y - 2x^2 \cos y}$.
c) Find the equation of the tangent to the curve at the point $(0, \frac{3\pi}{2})$ giving your answer in the form $y = mx + c$.

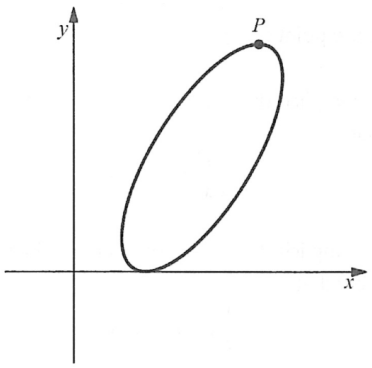
2.26 — Question 22. The equation of a curve is given by $x^2 - 2x \ln y = 1$.

a) Show that $(1, 1)$ lies on the curve.

b) Show that $\frac{dy}{dx} = \frac{y(x - \ln y)}{x}$.

c) Find the equation of the normal to the curve at the point $(1, 1)$ giving your answer in the form $y = mx + c$.

2.26 — Question 23. The curve $2(x - 5)^2 - 2xy + (y + 1)^2 = 1$ is shown in the diagram below.



a) Find the coordinates of the maximum point P of the ellipse.

b) Show that the distance between where the ellipse meets the x-axis and the point P is $k\sqrt{5}$ where k is an integer to be found.

2.26 — Question 24. The curve $y^2 + \ln x = k$, $x > 0$ where $k > 0$ passes through the point $(1, -\sqrt{2})$.

a) Find the value of k.

b) Find the exact coordinates at which the tangent to the curve is parallel to the y-axis.

2.26 — Question 25. The equation of a curve is given by $xe^y = 32$.

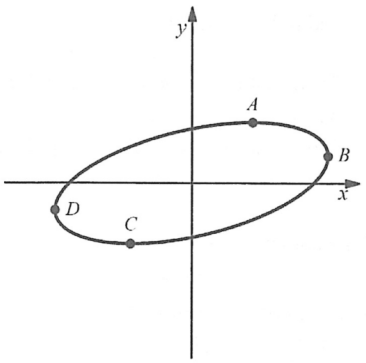
a) Show that the point $(2, 4\ln 2)$ lies on the curve.

b) Find the equation of the tangent line and the normal line at the point $(2, 4\ln 2)$.

c) Find the values where the equation of the tangent line and the normal line meet the x-axis.

d) Hence show that the triangle bounded by the tangent and normal line at $(2, 4\ln 2)$ and the x-axis has an area of $20(\ln 2)^2$.

2.26 — Question 26. Prerequisite knowledge for Q26: Domain and Range of Functions A curve is given by $x^2 - 2xy + 5y^2 = 20$ and shown on the diagram below.



The points A, B, C and D lie on the curve. The domain is such that the curve lies between the x-values of D and B. The range is such that the curve lies between the y-values of C and A. Find the coordinates of A, B, C and D.

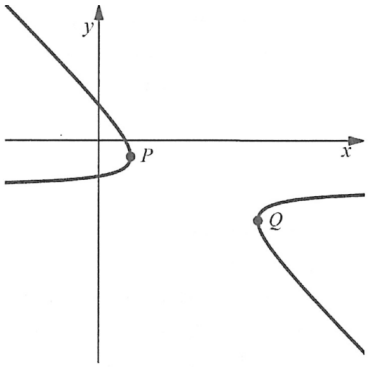
2.26 — Question 27. A curve is given by $xy^2 + x^3y = k$ where $k \neq 0$.

a) Show that the curve never crosses the x-axis.

b) Show that $\frac{dy}{dx} = \frac{-y^2-3x^2y}{2xy+x^3}$.

c) Given that a stationary point occurs when $x = -1$, find the value of k.

2.26 — Question 28. A curve has equation $3x + xy + y^2 = k$ where k is a positive integer such that $k < 9$.



There are two points $P(a, b)$ and $Q(5a, 5b)$ on the curve such that the tangents to the curve are parallel to the y-axis.

a) Find the values of a, b and k.

b) State the coordinates of the points P and Q.

2.26 — Question 29. A curve has equation $(x + y)^4 = 2x - y + 8$.

a) Show that the point $(-2, 3)$ lies on the curve.

b) Find the tangent line to the curve at the point $(-2, 3)$ in the form $ax + by = c$ where a, b and c are integers to be found.

c) Prove that the tangent line found in part b) does not intersect the equation again.

2.26 — Question 30. Prerequisite knowledge for Q30: Newton-Raphson Method The percentage of woodland cover in the UK, P, since 1965 may be modelled by the equation

$$P = 8.5 + \arctan(t - 31)$$

where t is the number of years since 1965.

a) Show that $\frac{d}{dx}(\arctan x) = \frac{1}{x^2+1}$.

b) Hence, find $\frac{dP}{dt}$.

c) Use Newton-Raphson's method starting at $t_1 = 30$ to find the year in which, according to the model, the percentage of woodland coverage is 9.95%.